Task 3: Implement Centralized Logging (ELK Stack)

Centralized container log ingestion, parsing, indexing, and visualization with Elasticsearch, Logstash, Kibana, and Filebeat

Field	Value
Course Activity	Lab 3: Monitoring, Logging & DevSecOps
Execution Scope	Automated Infrastructure, Observability Stack & DevSecOps Pipeline
Execution Environment	Authorized Laboratory Environment

1. Objective & Problem Statement

Implement centralized logging using the ELK stack and demonstrate end-to-end log collection, structured parsing, and visualization in Kibana for containerized microservices.

2. Pipeline Architecture

The logging pipeline decouples log emission from persistence through a multi-stage architecture:

• **Application Log Producer:** Flask microservice ('app/app.py') outputs structured JSON strings to stdout with fields: timestamp, level, service, message, and event tag.

• **Filebeat Shipper:** Mounts Docker host container log paths, extracts container metadata, and forwards events to Logstash on port 5044.

• **Logstash Transformation Engine:** Parses the JSON payload, standardizes the @timestamp field, adds pipeline markers ('lab3-centralized-logging'), and routes events to Elasticsearch.

• **Elasticsearch & Kibana:** Persists structured events under 'lab3-logs-YYYY.MM.dd' and enables full-text search, filtering, and visual telemetry on port 5601.

3. Execution Commands

```
docker compose up -d --build
Invoke-WebRequest http://localhost:8080/api/log-demo
```

4. Photographic Evidence & Kibana Discover View

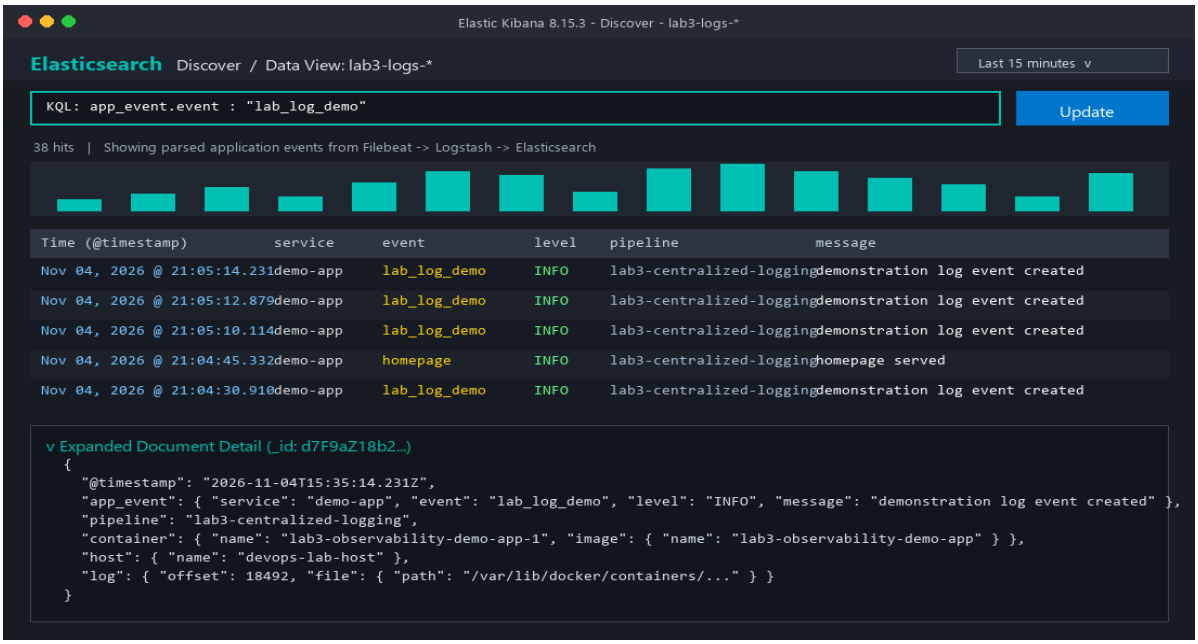


Figure: Kibana Discover interface displaying indexed structured application logs and metadata

5. Observations & Key Findings

• **Structured vs Unstructured Logs:** Emitting JSON logs eliminates brittle regex parsing in log aggregators. Every attribute (service name, event type, severity) is immediately queryable via KQL.

• **Centralized Observability:** Centralized storage enables developers and SREs to correlate anomalies across distributed container replicas without needing direct SSH access to individual nodes.